Week 1 Plan: Dataset and Ensemble

NEXA / Project Overwatch | Team: Chua, Jessy, Eileen, Eavan | Issued 27 Oct 2026

1. The goal in one sentence

By the end of day 7, the team hands over **one file,** `results/nexa_week1.pt`: a 5-model ensemble trained on the new combined dataset, with its class list, calibrated thresholds and the exact data and code it came from packed inside. Anyone who has this one file can run, evaluate or retrain the model without asking which checkpoint, config or dataset was used.

What goes inside the single .pt file

Key	Contents	Why it is inside
format_version	e.g. "nexa-week1-v1"	So a loader can refuse a file it does not understand
classes	Ordered class list (same order as the label columns)	The model's outputs mean nothing without it
signal	fs = 3.2 MS/s, window_len = 512	The input format the model expects
members	5 model state_dicts (seeds 1-5)	The ensemble itself; averaged at inference
thresholds	Per-class sigmoid thresholds from calibrate_thresholds.py --ensemble	Old thresholds are invalid after any retrain
git_commit	Code commit the members were trained with	Reproducibility
dataset	Combined-set version + list of shard manifests (name + SHA-256)	Proves exactly which data it learned from
metrics	Per-class recall / false-alarm on the synthetic test set and the bundle slice	The scorecard travels with the model

Today the ensemble is five separate files (results/ensemble_0.pt ... ensemble_4.pt) and src/infer.py loads them by file pattern. Packing them into one file needs a small packer script and a loader change in src/infer.py. Both are day-7 tasks for Eavan (see section 5).

Definition of done

- **results/nexa_week1.pt** exists, loads in src/infer.py, and produces a classification log on a test file.
- The combined dataset sits in `NEXA_data/combined/` with a QA report (window counts per class and SNR, thumbnails, no train/test leakage).
- ONNX exported from the same file and loading in the local server and UI with the new class list.
- Two scorecards: synthetic test set, and a hand-labelled slice (300-1,000 windows) of the partner's bundle.
- A week-2 fix list: the classes that drop most from synthetic to real.

2. Who generates what

Every person generates **every class in their row**. Classes marked week 2 are not in this week's ensemble.

Person	Classes	Count	Drive folder(s)	External data
Chua	**Jamming:** barrage, spot, tone, multi-tone, swept, chirp, pulsed, partial-band, reactive, DRFM deception (RGPO/VGPO). **Noise:** NOISE_FLOOR. **Pulse comms:** PPM (ADS-B), AFSK (APRS), SSR/IFF (1030 MHz). **Spread:** FHSS, DSSS, LoRa CSS. **Plus** YOLO wideband scenes (separate dataset).	17 + YOLO	jam/, pulse/, spread/, yolo/	RadDet (YOLO). Jammertest, Zenodo jamming: testing only
Jessy	**Radar, week 1:** plain pulse, LFM, Barker, polyphase Barker, Frank, Costas, FMCW, CW Doppler. **Week 2:** P1-P4, T1-T4, NLFM, stepped frequency. **Also week 1:** the impairment (realism) module.	8 now, 18 total	radar/	RadChar
Eileen	**Analog:** AM-DSB-WC, AM-DSB-SC, AM-SSB-WC, AM-SSB-SC, WBFM, NBFM, PM. **ASK:** OOK, 4ASK, 8ASK. **Multicarrier:** OFDM, LTE-like, 5G NR-like. **APSK:** 16APSK, 32APSK, 64APSK, 128APSK.	17	analog/, multicarrier/, apsk/	RadioML 2018, HisarMod, TorchSig (OFDM)
Eavan	**FSK:** 2FSK, 4FSK, 8FSK, GFSK, GMSK, MSK, CPFSK. **PSK:** BPSK, QPSK, OQPSK, 8PSK, 16PSK, 32PSK. **QAM:** 16QAM, 32QAM, 64QAM, 128QAM, 256QAM. **Also:** downloads external data once, model changes, mixtures, merge, training, packing the single .pt.	18 + merge + training	fskpsk/, mix/	RadioML 2018, HisarMod, CSPB

This week's ensemble therefore covers **60 classes** (17 + 8 + 17 + 18). Staggered PRI, jittered PRI and frequency-agile radar are **not** CNN classes: they only show up across many pulses, far longer than one 160 microsecond window, so they belong to a later pulse-train analyser.

3. The shared contract (agreed on day 1, frozen for the week)

Item	Rule
Window format	complex64, 512 samples at 3.2 MS/s (160 microseconds)
SNR grid	-10 to +20 dB in 2 dB steps (16 levels)
Generator signature	`generate(cls, n, snr_db, rng) -> (iq[n, 512], meta)`
Shard keys	**iq** complex64 (N, 512); **y** uint8 multi-hot (N, C) in configs/default.yaml class order; **snr_db** float32; **source** 'gen' or dataset name; **seed** int64
Shard name	`<group>_<split>_seed<seed>_<part>.npz`, at most 50,000 windows
Seed blocks	Chua 1000-1999, Jessy 2000-2999, Eileen 3000-3999, Eavan 4000-4999. Train, val and test use different seeds.
Manifest	One JSON per shard in `data/manifests/`: shard name, git commit, seed, split, class-list version, per-class counts, SHA-256. Committed to git.
External data cap	At most 30% of any class comes from an external dataset; the rest from our generators

4. How the data moves

Git holds the recipe, not the data. Arrays never go in git: the repo is public, GitHub rejects files over 100 MB, and some external datasets may not be redistributed. The arrays live in one shared Google Drive folder, synced to each laptop by Google Drive for desktop.

```
NEXA_data/
  external/     downloaded once by Eavan
  shards/
    jam/ pulse/ spread/ yolo/    Chua
    radar/                      Jessy
    analog/ multicarrier/ apsk/  Eileen
    fskpsk/ mix/               Eavan
  combined/   merged train/val/test set used for training
```

- **1. Generate locally.** Each person runs their generators on their own laptop and writes shards straight into their Drive folder (e.g. `G:\My Drive\NEXA_data\shards\radar\`). Drive uploads them by itself.
- **2. Push the recipe.** Each person pushes their generator code and the matching manifests to git.
- **3. Merge.** Eavan pulls, waits for Drive to finish syncing, checks every shard against its manifest SHA-256, builds the mixtures and writes `combined/`. YOLO scenes stay separate.
- **4. Train.** All five ensemble members train on the same combined version and the same commit.
- **Shortcut:** generators are seeded, so a shard can be regenerated from its manifest (commit + seed) instead of uploaded, provided code and library versions match.

Dataset size targets

Owner	Classes this week	Windows per class	Windows	Approx. size (complex64)
Chua	17	16 SNR x 1,000 = 16,000	272,000	1.1 GB
Jessy	8	16,000	128,000	0.5 GB
Eileen	17	16,000	272,000	1.1 GB
Eavan	18	16,000	288,000	1.2 GB
Eavan (mix/)	signal + jammer, signal + signal, 3-signal scenes	-	about 250,000	1.0 GB
Total classifier set	**60**		**about 1.2 million**	**about 4.9 GB**
Chua (yolo/)	Wideband scenes, 1-6 signals each, 10% noise-only	-	30k / 5k / 5k images	about 3 GB

Each person's uploads count against their own Drive quota (15 GB free), so every share above fits. Storing iq as float16 pairs would halve the size if space gets tight.

5. Day by day

Day	Chua	Jessy	Eileen	Eavan
1	Together, half a day: freeze the contract (section 3). Everyone installs Google Drive for desktop and the Python env.			Writes the class list into configs/default.yaml. Downloads RadioML 2018, HisarMod, RadChar, RadDet, CSPB into external/.
2	Jammer generators: barrage, spot, tone, multi-tone, swept, chirp, pulsed, partial-band, reactive, DRFM	Radar week-1 generators (extend src/generators/radar.py); hook up the RadChar loader	Analog AM x4, WBFM, NBFM, PM; ASK OOK, 4ASK, 8ASK	Change model head, per-class thresholds, ONNX export and UI class lists to 60 classes. Smoke-train on tiny fake data.
3	NOISE_FLOOR; pulse comms PPM, AFSK, SSR/IFF; spread FHSS, DSSS, LoRa	**Impairment module** + one test per impairment (frequency offset, clock ppm, phase noise, IQ imbalance, DC offset, fading, Doppler drift, AGC jumps, ADC bits). Hand over end of day.	Multicarrier OFDM, LTE-like, NR-like (TorchSig); APSK x4	FSK, PSK, QAM generators; converter for external data to 3.2 MS/s / 512 samples
4	YOLO scene generator that writes boxes automatically	Run every team generator through the impairment module; radar QA thumbnails	Convert external data for analog / ASK / APSK (30% cap)	Merge script (manifest + SHA-256 check) and mixture builder
4 end	**Generator freeze.** Anything not ready moves to week 2; the ensemble does not wait for it.			
5	Generate jam/, pulse/, spread/ shards + manifests; generate yolo/ scenes	Generate radar/ shards + manifests	Generate analog/, multicarrier/, apsk/ shards + manifests	Generate fskpsk/; merge everything into combined/; QA report
5 end	**Dataset handoff:** combined/ is final. No changes after this point.			
6	Train ensemble **seed 3**	Train **seed 4**; hand-label the bundle slice with Eileen	Train **seed 5**; hand-label the bundle slice with Jessy	Train **seeds 1 and 2**
7	YOLO dataset QA (box overlays, counts)	Bundle-slice scorecard; synthetic-vs-real gap	Synthetic test-set scorecard; week-2 fix list	calibrate_thresholds.py --ensemble; **pack results/nexa_week1.pt**; loader change in src/infer.py; ONNX export; activate in local server

Rows "4 end" and "5 end" are hard deadlines for the whole team.

Training rules (so five people produce one ensemble)

- Everyone trains from the **same git commit** and the **same combined/ version**; write both into the run log.
- Seeds 1-5 are fixed in advance (table above). Do not re-run a seed with a different value.
- Save a checkpoint to Drive every epoch; if a Colab session dies, resume from it.
- Copy the finished member to `NEXA_data/members/seed<k>.pt`. Eavan packs all five into the single file on day 7.

6. Risks and what to do

Risk	Effect	What we do
A generator is late	Delays the whole dataset	Day-4 freeze: the class moves to week 2
Data looks too clean	Big drop on real data	Impairment module (Jessy, day 3) is applied to every generator
Shard half-uploaded or changed	Silent bad training data	Merge checks every SHA-256 against its manifest
Members trained on different data/code	Ensemble is not one model	Same commit + combined/ version recorded in each run log and in the .pt
Colab session cut off	Lost hours	Checkpoint every epoch; resume
Class count changes from 8 to 60	Old thresholds, ONNX, UI and retrain gate break	Eavan changes them on day 2 and smoke-tests
Retrain gate compares old 8-class vs new 60-class model	Gate cannot compare	Treat week 1 as a new model line; gate applies from week 2
Bundle is unlabelled	No real accuracy number	Hand-label a 300-1,000 window slice (day 6)
External licences (RadChar is CC BY-NC; RadioML unconfirmed)	Cannot ship in a commercial product	Research and exhibition use only; check with the partner

7. End-of-week checklist

Done?	Item	Owner
[ ]	Contract frozen; class list in configs/default.yaml	All / Eavan
[ ]	External data in NEXA_data/external/	Eavan
[ ]	Impairment module + tests merged	Jessy
[ ]	All shards + manifests pushed (jam, pulse, spread, radar, analog, multicarrier, apsk, fskpsk)	Everyone
[ ]	combined/ written, every SHA-256 verified, QA report	Eavan
[ ]	YOLO scenes 30k / 5k / 5k with box QA	Chua
[ ]	Five members trained (seeds 1-5)	Eavan x2, Chua, Jessy, Eileen
[ ]	Thresholds recalibrated on the ensemble	Eavan
[ ]	**results/nexa_week1.pt** packed and loading in src/infer.py	Eavan
[ ]	ONNX exported; local server and UI show the 60 classes	Eavan
[ ]	Scorecards: synthetic test set + bundle slice	Eileen, Jessy
[ ]	Week-2 fix list	Eileen